

Product Carbon Footprint Study Report

Product Name: vacuum cleaner

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: vacuum cleaner

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 item vacuum cleaner

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 item vacuum cleaner.

2. System Boundary

vacuum cleaner production in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of vacuum cleaner produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **38.366 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

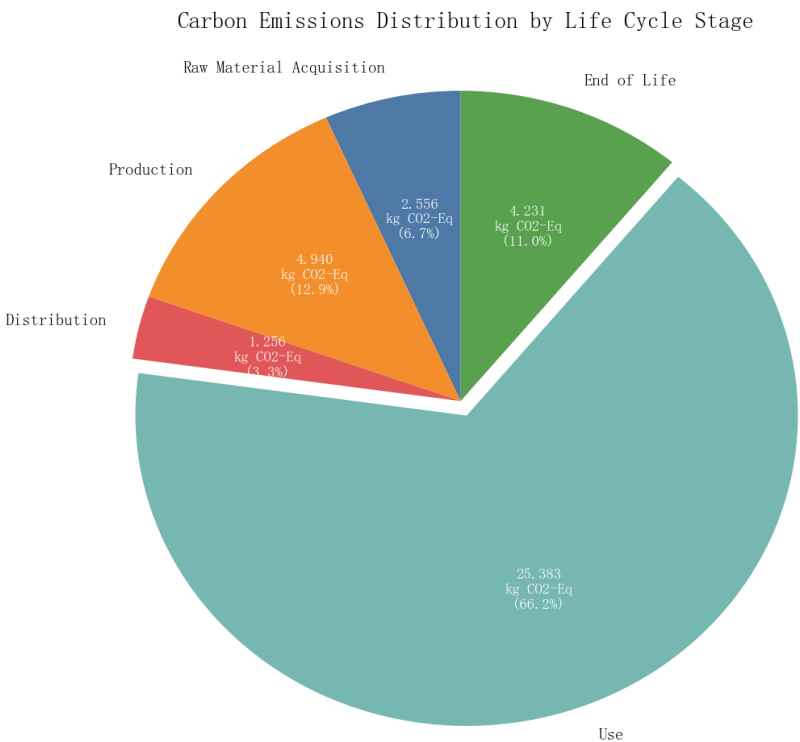


Table 1: Carbon Emissions by Life Cycle Stage

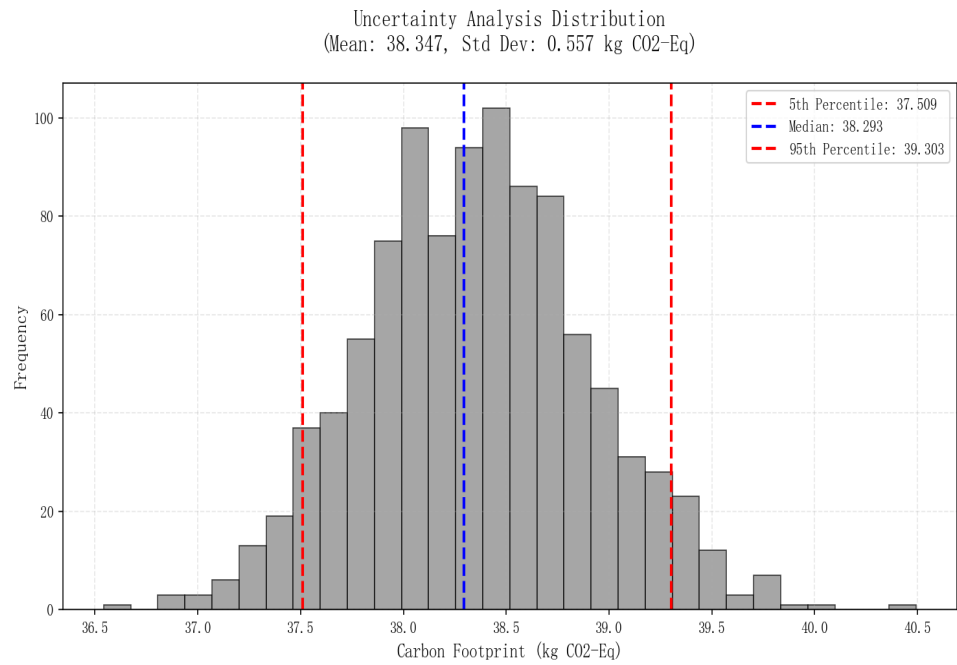
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	2.556	6.66
Production	4.940	12.88
Distribution	1.256	3.27

Use	25.383	66.16
End of Life	4.231	11.03
Total	38.366	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO2-Eq)
Mean	38.347
Standard Deviation	0.557
Coefficient of Variation (CV)	1.45%
Median	38.293
5th Percentile	37.509
95th Percentile	39.303

VI. Interpretation of Results

1. Description of Results

The vacuum cleaner's total carbon footprint of 38.366 kg CO₂-Eq represents a moderate impact level for a household appliance, primarily driven by operational energy demands over its service life. The lifecycle stage contributions reveal a highly skewed distribution, with the use phase accounting for 66.2% (25.383 kg CO₂-Eq) of emissions, followed by production at 12.9% and end-of-life at 11.0%. Raw material extraction and distribution contribute modestly at 6.7% and 3.3%, respectively, highlighting that indirect emissions from electricity consumption during consumer use dominate the overall profile.

This characteristic underscores the product's classification as a use-intensive item where operational efficiency outweighs upstream manufacturing burdens. Major emission sources are concentrated in the use stage, reflecting typical grid electricity mixes, while production and disposal phases present secondary hotspots linked to material processing and waste treatment. The relatively low distribution impact suggests efficient logistics, but opportunities exist to address the dominant use-phase emissions through design interventions.

2. Statement of Assumptions and Limitations

Key assumptions in this study include standardized usage patterns of 50 hours annually over a 10-year lifespan, reliance on average European electricity grid emission factors, and generic material compositions for components such as plastics and motors. End-of-life scenarios assume partial recycling rates consistent with EU WEEE directives, while transport distances are modeled using representative regional averages.

Limitations arise from data sources that predominantly utilize secondary databases like Ecoinvent, which may not capture proprietary supplier-specific processes or regional variations in energy mixes, potentially leading to over- or under-estimation of impacts by 10-20%. Data quality for rare materials or emerging technologies remains moderate due to temporal gaps in inventory datasets.

System boundary restrictions encompass a cradle-to-grave approach limited to the product's direct lifecycle, excluding indirect effects such as consumer transport to retail, packaging end-of-life beyond the product, and broader societal infrastructure impacts associated with electricity generation.

3. Suggestions for Improvement

- Target the use phase by redesigning the motor for 20-30% higher energy efficiency through brushless DC technology, directly reducing the 25.383 kg CO₂-Eq contribution.
- Implement modular design for easier component replacement and increased recycled content in plastics to lower raw material and production emissions by at least 15%.
- Shift to renewable energy-powered manufacturing facilities and optimize packaging to cut distribution impacts while exploring take-back programs to improve end-of-life recycling rates.
- Enhance data quality by conducting primary data collection from key suppliers for material-specific inventories and performing sensitivity analyses on usage scenarios.

- Pursue product-as-a-service models to extend lifespan and enable real-time efficiency monitoring, providing feasible pathways for overall footprint reduction below 30 kg CO₂-Eq.

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